

2021-Ph-H1-Q17

Section: Particles and Waves

Topic: Interference

Which setup shows destructive interference?

Solution:

Destructive interference occurs when path difference =

$$(n + \frac{1}{2})\lambda$$

This results in a crest meeting a trough.

Answer: D

Guidance for Students:

In Young's slits or any interference pattern, bright = constructive, dark = destructive. Look for half-wavelength path differences.

Revision Tip:

Use:

- Constructive: $\Delta d = n\lambda$
- Destructive: $\Delta d = (n + \frac{1}{2})\lambda$